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// StudentManagementSystem.java
// Main class - runs a menu-driven console
app to manage students.
// Demonstrates: Collections (ArrayList),
Loops, Switch-case, Exception handling

import java.util.ArrayList;
import java.util.Scanner;

public class StudentManagementSystem {

    // ArrayList to store all students
    (acts as our "database")
    private static ArrayList<Student>
studentList = new ArrayList<>();
    private static Scanner scanner = new
Scanner(System.in);
    private static int nextId = 1; // auto-
incrementing ID

    public static void main(String[] args)
{
    boolean running = true;

    while (running) {
        printMenu();
        int choice = readInt("Enter
your choice: ");

        switch (choice) {
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        case 1:
            addStudent();
            break;
        case 2:
            viewAllStudents();
            break;
        case 3:
            updateStudent();
            break;
        case 4:
            deleteStudent();
            break;
        case 5:
            searchStudent();
            break;
        case 6:
            running = false;

System.out.println("Exiting... Goodbye!");
            break;
        default:

System.out.println("Invalid choice. Please
try again.\n");
    }
    }
    scanner.close();
}

private static void printMenu() {
    System.out.println("==== Student
Management System =====");
    System.out.println("1. Add
Student");

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        System.out.println("2. View All
Students");
        System.out.println("3. Update
Student");
        System.out.println("4. Delete
Student");
        System.out.println("5. Search
Student by ID");
        System.out.println("6. Exit");
    }

    // ----- CRUD Operations -----
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    private static void addStudent() {
        System.out.print("Enter name: ");
        String name = scanner.nextLine();
        int age = readInt("Enter age: ");
        System.out.print("Enter course: ");
        String course = scanner.nextLine();

        Student student = new
Student(nextId, name, age, course);
        studentList.add(student);
        nextId++;

        System.out.println("Student added
successfully!\n");
    }

    private static void viewAllStudents() {
        if (studentList.isEmpty()) {
            System.out.println("No students
found.\n");

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        return;
    }
    System.out.println("---- All
Students ----");
    for (Student s : studentList) {
        System.out.println(s);
    }
    System.out.println();
}

private static void updateStudent() {
    int id = readInt("Enter the ID of
the student to update: ");
    Student student =
findStudentById(id);

    if (student == null) {
        System.out.println("Student not
found.\n");
        return;
    }

    System.out.print("Enter new name
(leave blank to keep \"" +
student.getName() + "\"): ");
    String name = scanner.nextLine();
    if (!name.isEmpty()) {
        student.setName(name);
    }

    int age = readInt("Enter new age
(enter 0 to keep \"" + student.getAge() +
"\"): ");
    if (age != 0) {

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        student.setAge(age);
    }

    System.out.print("Enter new course
(leave blank to keep \"" +
student.getCourse() + "\"): ");
    String course = scanner.nextLine();
    if (!course.isEmpty()) {
        student.setCourse(course);
    }

    System.out.println("Student updated
successfully!\n");
}

private static void deleteStudent() {
    int id = readInt("Enter the ID of
the student to delete: ");
    Student student =
findStudentById(id);

    if (student == null) {
        System.out.println("Student not
found.\n");
        return;
    }

    studentList.remove(student);
    System.out.println("Student deleted
successfully!\n");
}

private static void searchStudent() {
    int id = readInt("Enter the ID to

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search: ");
        Student student =
findStudentById(id);

        if (student == null) {
            System.out.println("Student not
found.\n");
        } else {
            System.out.println("Found: " +
student + "\n");
        }
    }

    // ----- Helper Methods -----

    private static Student
findStudentById(int id) {
        for (Student s : studentList) {
            if (s.getId() == id) {
                return s;
            }
        }
        return null;
    }

    // Safely reads an integer from the
user, handling invalid input
    private static int readInt(String
prompt) {
        System.out.print(prompt);
        while (!scanner.hasNextInt()) {
            System.out.print("Please enter
a valid number: ");
            scanner.next();

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    }  
    int value = scanner.nextInt();  
    scanner.nextLine(); // consume  
    leftover newline  
    return value;  
  }  
}
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